Laboratório 07

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
setwd("C:/Users/jvald/Downloads/ME315")

library(RSQLite)
```

```
Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3
```

```
library(DBI)

conn <- dbConnect(SQLite(),"uwmadison.sqlite3")
dbListTables(conn)
```

```
 [1] "course_offerings"    "courses"             "grade_distributions"
 [4] "instructors"         "rooms"               "schedules"
 [7] "sections"            "subject_memberships" "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation=‘STAT’);

```
dbGetQuery(conn, "SELECT COUNT(DISTINCT instructors.id) AS quantidade
FROM subjects
JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
JOIN teachings
```

```
    ON sections.uuid = teachings.section_uuid
JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'")
```

```
  quantidade
1        379
```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
professores <- dbGetQuery(conn, "
SELECT instructors.name, AVG((4*gd.a_count + 3.5*gd.ab_count + 3*gd.b_count + 2.5*gd.bc_count

FROM grade_distributions AS gd
JOIN subject_memberships AS sm ON gd.course_offering_uuid = sm.course_offering_uuid
JOIN subjects AS s ON sm.subject_code = s.code
JOIN sections ON gd.course_offering_uuid = sections.course_offering_uuid
AND gd.section_number = sections.number
JOIN teachings ON sections.uuid = teachings.section_uuid
JOIN instructors ON teachings.instructor_id = instructors.id
WHERE s.abbreviation = 'STAT'
GROUP BY instructors.id, instructors.name
ORDER BY gpa_medio")
head(professores)
```

```
                  name gpa_medio
1    CHRISTINE SORKNESS        NA
2 NATALIA DE LEON GATTI        NA
3        GUILHERME ROSA        NA
4       MARIAN R FISHER        NA
5       MICHELLE HARRIS        NA
6  MICHAEL RENE KOSOROK        NA
```

```
dbWriteTable(conn, "professores", professores, temporary = TRUE)
```

Quem é o professor mais difícil?

```
dbGetQuery(conn, "SELECT name, gpa_medio
FROM professores
WHERE gpa_medio IS NOT NULL
ORDER BY gpa_medio
LIMIT 1")
```

```
            name gpa_medio
1 JAMES D KUELBS    2.6375
```

Quem é o professor mais fácil?

```
dbGetQuery(conn, "SELECT name, gpa_medio
FROM professores
WHERE gpa_medio IS NOT NULL
ORDER BY gpa_medio DESC
LIMIT 1")
```

```
                        name gpa_medio
1 SRIKANTHMADHAVAN ARAVAMUTHAN         4
```

```
disciplinas <- dbGetQuery(conn, "SELECT courses.number, courses.name, AVG(( 4*gd.a_count + 3
FROM grade_distributions AS gd
JOIN subject_memberships AS sm
ON gd.course_offering_uuid = sm.course_offering_uuid
JOIN subjects AS s
ON sm.subject_code = s.code
JOIN course_offerings
ON gd.course_offering_uuid = course_offerings.uuid
JOIN courses
ON course_offerings.course_uuid = courses.uuid
WHERE s.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.number, courses.name
ORDER BY gpa_medio")
head(disciplinas)
```

```
  number                                                          name gpa_medio
1    698                                                Directed Study        NA
2    546 Practicum in Clinical Trial Data Analysis and Interpretation        NA
3    990                                                      Research        NA
4    768              Statistical Methods for Medical Image Analysis        NA
5    699                                                Directed Study        NA
6    681                                          Senior Honors Thesis        NA
```

```r
dbWriteTable(conn, "disciplinas", disciplinas, temporary = TRUE)
```

Qual é a disciplina mais difícil?

```r
dbGetQuery(conn, "SELECT name, gpa_medio
FROM disciplinas
WHERE gpa_medio IS NOT NULL
ORDER BY gpa_medio
LIMIT 1")
```

```
                                     name gpa_medio
1 Introduction to the Theory of Probability  2.891568
```

Qual é a disciplina mais fácil?

```r
dbGetQuery(conn, "SELECT name, gpa_medio
FROM disciplinas
WHERE gpa_medio IS NOT NULL
ORDER BY gpa_medio DESC
LIMIT 1")
```

```
                                                  name gpa_medio
1 Nonparametric Statistics and Machine Learning Methods         4
```

4. Desconecte do banco de dados.

```r
dbDisconnect(conn)
```

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 19:08:24 -03"
```